# Hyunin Lee

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## Education

**University of California, Berkeley** — CA, United States
*Ph.D. in Mechanical Engineering / Advisor: Dr.Somayeh Sojoudi* — *Aug. 2022 – May. 2027 (expected)*

**Seoul National University** — Seoul, Rep.of.Korea
*B.S in Mechanical Engineering; summa cum laude* — *Mar. 2015 – Feb. 2022*

## Research Statement

My research aims to develop AI systems capable of solving non-verifiable problems far beyond human supervision, with a current focus on recursive self-improving multi-agent systems [P3, P4]. My previous work has explored training agents in time-varying environments [C2, C3] and AI safety [C4, C5]. My research has been published at ICML, ICLR, and NeurIPS, including an oral presentation.

## Publications / C: Conference, J: Journal, P: Preprint

[**P4**] Recurisve Self-Improvement Thougth Multi-Agent Self-Superivision.
**H. Lee**, J. Xu, J. Seely, D. Lee, S. Sojoudi, M. Zaharia and Y. Tang. *Preprint.* 2026. [**pdf/blog**]

[**P3**] Recurisve Harness Self-Improvement.
**H. Lee**, J. Xu, J. Seely, D. Lee, M. Zaharia and Y. Tang. *Preprint.* 2026. [**pdf/blog**]

[**P2**] Cross-attention Secretly Performs Orthogonal Alignment in Recommendation models.
**H. Lee**, Y. Zhang, H. Nguyen, X. Liu, N. Park, C. Jung, Y. Wang, Z. Wang, S. Sojoudi, X. Feng. *Under review.* 2026. [**pdf/blog**]

[**P1**] A Prospect Theoretic Rationality: Loss Aversion Can Accelerate Reinforcement Learning
**H. Lee**, C. Park, S. Sojoudi, N. Mehr. *preprint.* 2026. [**pdf**]

[**C5**] Position: AI Safety Must Embrace an Antifragile Perspective.
M. Jin, **H. Lee**. *ICML.* 2025. [**pdf**]

[**C4**] A Black Swan Hypothesis: The Role of Human Irrationality in AI Safety.
**H. Lee**, C. Park, D. Abel, M. Jin. *ICLR.* 2025 & *ICLR 2025 Advances in Financial AI Workshop* (Oral, Top 1.2%). [**pdf**]

[**C3**] Pausing Policy Learning in Non-stationary Reinforcement Learning.
**H. Lee**, M. Jin, J. Lavaei, and S. Sojoudi. *ICML.* 2024. (Oral, Top 1.2%) [**pdf/codes/talk**]

[**J3**] Policy-based Primal-Dual Methods for Concave CMDP with Variance Reduction.
D. Ying, M. Guo, **H. Lee**, Y. Ding, J. Lavaei, and Z. Shen. *JAIR.* 2025. [**pdf /codes**]

[**C2**] Tempo Adaptation in Non-stationary Reinforcement Learning.
**H. Lee**, Y. Ding, J. Lee, M. Jin, J. Lavaei, and S. Sojoudi. *NeurIPS.* 2023 [**pdf/codes/slides**]

[**J2**] Beyond Exact Gradients: Convergence of Stochastic Soft-Max Policy Gradient Methods with Entropy Regularization.
Y. Ding, J. Zhang, **H. Lee**, and J. Lavaei. *IEEE TAC.* 2025 [**pdf**]

[**C1**] Initial State Interventions for Deconfounded Imitation Learning.
S. Pfrommer, Y. Bai, **H. Lee**, and S. Sojoudi. *IEEE CDC.* 2023. [**pdf**]

[**J1**] Explainable Deep Learning Model for EMG Based Finger Angle Estimation Using Attention.
**H. Lee**, D. Kim, and Y. Park. *IEEE TNSRE.* vol. 30, pp. 1877-1886 2022. [**pdf/codes**]

## Work Experience

**SakanaAI** May 2026 – Aug 2026
*Research Intern* *RnD team — Tokyo, Japan*

- Researched recursive self-improving LLM-based multi-agent systems [**P3**, **P4**]. (Intern Manager: Yujin Tang)

**Meta** May 2025 – Oct 2025
*Research Scientist Intern* *Ranking AI Research Team — Sunnyvale, CA*

- Researched on transformer-based **multi-modal recommendation model**, enhancing ranking performance in sparse data setting. (Intern manager: Yong Zhang, Xue Feng)
- Proposed a novel cross-attention mechanism—orthogonal alignment—achieving 1.4x improvement in **scaling law efficiency**, validated on both Meta ranking model and public baselines. [**p1**]

**OpenAI** Mar 2025 – Aug 2025
*Research Associate* *Human Data (Research) Team — Remote*

- Engaged with the safety and preparedness team to evaluate an AI agent's ability to replicate ML research and to create a comprehensive rubric that defines objective success criteria for accurately reproducing given ML papers. (Manager: Benjamin Kinsella)

**University of California, Berkeley** Aug 2022 – Present
*Graduate Student Researcher* *Advisor: Dr. Somayeh Sojoudi*

- Research on **non-stationary reinforcement learning** and optimization under distributional shift. **[p1, p2]**
- Research on **Human-AI alignment** focusing on ensuring **safe AI systems** in the presence of unpredictable or non-standard human behaviour. **[p1, p2, p3]**
- Research on optimal workflow control of LLM-based Multi-agent.

**OUTTA** **[Homepage]** **[LinkedIn]** Aug 2021 – Present
*Co-Founder* *Seoul, South Korea*

- Deliver online AI courses to **800+ students** annually across South Korea.

**Knowledge AI** Jul 2021 – Jul 2022
*Machine Learning Engineer* *Boston, MA*

- Implemented a **Bayesian inference algorithm** in Python to quantify students' mastery of mathematics topics.
- Developed a question-recommendation deep-learning system for an online math-learning platform, boosting student performance.

**Seoul National University** Mar 2021 – Nov 2021
*Undergraduate Research Intern* *Soft Robotics & Bionics Lab*

- Designed an **attention-based sequential decision-making algorithm** in Python to predict finger-joint angles from forearm EMG signals, improving accuracy by more than 10%.

**Seoul National University** Sep. 2020 – Jun. 2021
*Undergraduate Research Intern* *Robot Learning Lab*

- Develop **deep generative Q learning algorithm** to reconstruct a reward kernel using Python **[pdf]** **[video]**

## Academic Service

**Reviewer** ICLR 2024-2027, ICML 2024-2027, NeurIPS 2024-2027, RLC 2025, AAAI 2025-2027
**Program Chair Committee** AAAI 2025 - 2027

## Teaching Experience

**Graduate Student Instructor, Statistics and Data Science for Engineers** Spring, Fall 2025 / Spring 2026

**Graduate Student Instructor, Dynamic Systems and Feedback** Fall 2024

**Teaching Tutor, Math and Physics for Freshman** Spring 2019 / Spring 2020

**Teaching Assistant, Dynamic** Fall 2019

**Teaching Assistant, Mechanical Product Design** Fall 2020

## Grants and Honors

**William C Webster Graduate Fellowship** | *Mechanical Engineering Department* — Fall 2026

**Graduate Division Block Gran** | *Mechanical Engineering Department* — Summer 2026

**Berkeley Summer Research Fellowship** | *Mechanical Engineering Department* — Summer 2024

**NeurIPS scholar award** | *Conference on Neural Information Processing Systems* — Dec. 2023

**Kwanjeong Abroad Scholarship** | *Kwanjeong Educational Foundation* — Fall 2022 – Present

**Berkeley Fellowship for Graduate Study** | *Graduate Division* — Fall 2022 – Spring 2023

**National Science & Technology Scholarship** | *Korea Student Aid Foundation* — Spring 2017, Fall 2019, Spring 2020, Fall 2020

**Certificate of Appreciation (OUTTA)** | *Dean, college of Engineering, Seoul National University* — Jun. 2021

**Scholarship to Academic Excellence** | *Seoul National University* — Spring 2015, Fall 2015, Spring 2016, Fall 2016

## Graduate courses

*Specialization: Non-convex Optimization & Reinforcement Learning*

**Theoretical statistics I, II**, **Probability Theory I, II**
**Convex Optimization** (convex optimziation, robust optimization)
**Mathematical Programming II** (Advanced optimization theory, non-convex optimization)
**Advanced control system I** (canonical state-space representation forms, Lyapunov stability, LQR control)
**Experiential advanced control design I, II** (model predictive control, kalman filter)
**Linear System, Nonlinear System**

## Technical Skills

**Languages**: Python (Advanced), MatLab (Advanced), C++
**Software library, platform** : Pytorch (Advanced), Tensorflow. Gurobi (Advanced), CPLEX (Advanced)